ERS

RADIANODE ENGINEERING READINESS SERIES

ERS-2026-003

Reproducibility Statement

Companion to ERS-2026-003 v1.0, "Why Carrier-Board Module-Current Measurements Can Be Misleading"

FIELD	VALUE
Statement version	1.0, issued 16 October 2026
Applies to	ERS-2026-003 v1.0, October 2026
Publication SHA-256	`6c12019b58907b6fc6f7894f809283d377f6dc6577e4cc3bae7952c0461d0321`
Repository	github.com/radianode/ers-2026-003
Bench session	ERS003-BENCH-01, 14 October 2026; session log SHA-256 `312fcd32c371ac9df1c0bc5f27975347d8f8382a74394df2b37c435087aee538`
Waveshare HAT re-measurement	15 October 2026; two recordings (E31, E32)
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1 Purpose

This statement documents the evidence files, analysis methods and reference values needed to reproduce the measurements and figures in ERS-2026-003 v1.0. Source evidence, derived data and comparator documents are assigned identifiers (E, X and D respectively). Section 10 lists the corresponding filenames and SHA-256 hashes. Evidence that depends on operator readings or statements is distinguished from results that can be recalculated directly from files.

2 Repository contents

CONTENT	ITEMS	NOTE
Raw captures, DIP40CORE-A (14 Oct 2026)	E18 to E23, E27: .ppk2 state files and CSV exports for runs 1 to 3 and the UART test	Hosted in a Zenodo dataset, not in the GitHub repository, because four of the files exceed GitHub's 100 MB file limit (E21 alone is 5,087,288,923 bytes). Dataset DOI: 10.5281/zenodo.23399519 (https://doi.org/10.5281/zenodo.23399519). Each file's SHA-256 is in section 10. E20 is the application's saved state of the same run 3 recording as E21
Raw captures, Waveshare HAT (15 Oct 2026)	E31, E32: CSV exports of recordings 1 and 2 (300.00 s, 100 kS/s)	Hosted in the same Zenodo dataset; each file is above GitHub's 100 MB limit
Raw capture, Waveshare HAT (26 Aug 2026)	E12 (10.000 s, 100 kS/s)	From the ERS-2026-002 working archive; held in the GitHub repository
Screenshots and transcripts	E2 to E6, E17, E24 to E26, E30, E33, E34	Application views, AT transcripts
Photographs	E13, E14, E16, E28, E29, E35 to E37	Figures 4 and 6 of the report redact the modules' unit identifiers (serial number where marked, IMEI, QR code). Figure 4 shows E13 and crops of E36 and E37; Figure 6 shows two crops of E16, the marking enlarged three times. The repository holds the unredacted photographs by deliberate decision; the redactions apply to the report figures only
Session record	E15, frozen at release	Every step, reading, deviation and error of the 14 October session
Derived data	X1 to X4: per-second summaries	Computed by the analysis script below
Scripts	scripts/ers003_analysis.py; scripts/figures/	Analysis and figure generation
Comparator documents	D1 to D8: version, table and SHA-256 only	Vendor documents are not redistributed. D6 and D7 were supplied by the manufacturer

3 Equipment and conditions

ITEM	RECORD
Current instrument	Nordic Power Profiler Kit II, serial D439F20BA129, Power Profiler v4.4.1; source-meter mode, 3,800 mV, 100,000 samples/s (E17)
Multimeter	MS8268, uncalibrated; DC volts for STATUS, VDD-EXT, VIN and VBAT; diode mode for the VBAT-to-VIN path
Device under test	DIP40CORE-A BG95 V1.1 carrying Quectel BG95-M3, SKU BG95M3LA-64-SGNS, firmware BG95M3LAR02A03; no antenna, no SIM

ITEM	RECORD
Connections	ERS-2026-003 Figure 8: PPK2 to pins 39 and 40; VIN (pin 2) unconnected; UART pins 29/30 through a TI TXS0108E (VA from VDD-EXT pin 32, OE tied to VA, VB from the Pi 3.3 V) to a Raspberry Pi 4 at 115200 8N1. Both UART paths confirmed working by the AT exchange (E24, E25). The setup photographs (A1_4_setup_overview_1/2, A1_4_level_converter) do not resolve individual connections, and the open VIN was not confirmed by a continuity check
Waveshare HAT re-measurement	15 October 2026: same instrument and settings (E33, E34). SIM7080G Cat-M/NB-IoT HAT with a SIM fitted and no antenna; PPK2 VOUT and GND to the HAT's VBAT and GND header pins, no other connections; USB empty, not on a Pi, P2 bridging 5V to VCC, PWR-to-P7 open (E35 to E37)
Ambient	14 October: 27 °C at the start and end of the session (Taidacent SHT30, calibration unknown). 15 October: not recorded

4 Data formats

CSV export (E21, E23, E27, E12, E31, E32). Header `Timestamp(ms),Current(uA),D0-D7`; one row per sample at 10 µs spacing.

.ppk2 state file (E18 to E20, E22), as read by the analysis script for Power Profiler v4.4.1. A zip archive holding `metadata.json` (for example `{"metadata":{"samplesPerSecond":100000,"startSystemTime":1792006398604},"formatVersion":2}`), `session.raw` and `minimap.raw`. `session.raw` is a sequence of 6-byte records: little-endian float32 current in µA, then uint16 digital-channel bits. `startSystemTime` is the recording start in Unix milliseconds (UTC); the session times in the report add one hour (WAT). This layout is not taken from Nordic documentation; it was established by reading the file and confirmed by matching the first samples of E20 with E21 exactly.

5 Analysis

All DIP40CORE-A statistics come from per-second bins of 100,000 samples (mean, minimum, maximum, count). Segment and whole-run means are count-weighted means of the bins, computed with `scripts/ers003_analysis.py` (Python 3, NumPy; pandas for CSV input).

The commands below are run from the repository's `scripts/` directory. Download the required source files from the dataset before execution and supply their local paths where necessary. The output filenames shown in the commands are used by subsequent analysis steps.

STEP	COMMAND
Per-second summary from a state file	`python3 ers003_analysis.py ppk2 A2_run2.ppk2 X2.csv`
Per-second summary from a CSV export	`python3 ers003_analysis.py csv A2_run3_off.csv X3.csv`
Segment means	`python3 ers003_analysis.py stats X3.csv 0:600 600:1200 1200:1800 0:1801`
Short events	`python3 ers003_analysis.py events A2_run3_off.csv 1300 2000`
Waveshare HAT windows and Figure 2	`python3 figures/fig_board1_raw.py E32.csv E12.csv fig2.png`

For the Waveshare HAT recordings, the PWRKEY transitions are located where the 1 ms mean first and last exceeds 5 mA; the off-state windows exclude 1 s either side of each transition. Run with E31 in place of E32 for recording 1.
Tested software versions. Python 3.10.12, NumPy 2.2.6, pandas 2.3.3, Matplotlib 3.10.9 (X1 from E18, reproduced exactly; Figure 2 and the Waveshare HAT values in section 6). Python 3.11.15, NumPy 2.4.4, Matplotlib 3.10.9, Pillow 12.2.0 (segment means from X2 to X4; Figures 4 and 9 to 12); these steps read no CSV export, so pandas was not used in this environment.
The per-second summaries X1 to X4 were produced on 14 October 2026 by earlier versions of the same procedure. The script reproduces X1 exactly from E18 (checked) and returns the segment means below from X2 to X4.

6 Values a correct reproduction returns

Interval notation: seconds are counted from zero, and [a, b) denotes the whole seconds a to b − 1, as computed by the `stats` command. For example, [0, 433) is the 433 complete seconds before the transition second at 433 s.

QUANTITY	INPUT	VALUE
Run 3: samples; mean; standard deviation; max	E21 (or E20)	180,000,001; 1538.82 µA; 36.03 µA; 1809.87 µA
Run 3: means of seconds [0, 600), [600, 1200), [1200, 1800)	X3	1539.01; 1538.67; 1538.78 µA
Run 3: range of 1 s means	X3	1536.34 to 1543.96 µA
Run 3: samples below 1300 µA	E21	45, in three runs of 15 at 157.578, 158.874 and 175.360 s, each reading 0.2 µA
Run 2: recording start (metadata)	E19	2026-10-14 19:33:18.604 UTC
Run 2: means of seconds [0, 433), [434, 960), [1053, 1310), [1310, 1801)	X2	26260.2; 1539.0; 26240.8; 1538.8 µA
Run 2: largest per-second maximum	E19	1,681,909 µA at second 960
UART test: means of seconds [0, 76), [79, 169), [172, 300)	X4	1539.39; 1538.77; 1539.17 µA
UART test: whole-record mean; maximum	E23	1539.20 µA; 25,906 µA at 77.556 s
Waveshare HAT, 26 Aug: mean; maximum; samples	E12	2.062 mA; 2.148 mA; 1,000,001
Waveshare HAT recording 2: power-on; power-down	E32	212.486 s; 279.294 s
Waveshare HAT recording 2: mean before power-on; after power-down	E32	1.8941 mA (211.5 s); 1.8856 mA (19.7 s)
Waveshare HAT recording 2: on-state mean; maximum	E32	23.59 mA; 221.2 mA
Waveshare HAT recording 1: power-on; power-down	E31	80.164 s; 243.847 s
Waveshare HAT recording 1: mean before power-on; after power-down	E31	1.8823 mA (79.2 s); 1.8721 mA (55.2 s)
Waveshare HAT recording 1: samples below 1300 µA	E31	15, at 247.921 s, each reading 0.2 µA

7 Figures

REPORT FIGURE	SCRIPT	INPUT
1 PPK2 operating mode	figures/fig_mode.py (crops and annotations)	E30, E5, E6
2 Waveshare HAT header current	figures/fig_board1_raw.py	E32, E12
3 ERS-2026-002 Figure 4	Extracted panel from ERS-2026-002, Figure 4 (E8); no separate generation script	E8
4 Waveshare HAT measurement configuration	figures/fig_board1_photos.py (crops; IMEI and QR code redacted)	E13, E36, E37
5, 7 VBAT nets (redrawn)	figures/figs4.py	D5, D6 (drawn from, not reproduced)

REPORT FIGURE	SCRIPT	INPUT
6 Module marking	figures/fig_marking.py (two crops; marking enlarged three times; serial number, IMEI and QR code redacted)	E16
8 DIP40CORE-A connections	figures/fig_wiring.py	E15
9 Off state 30 min (run 3), 10 UART test, 11 Operation to power-down (run 2), 12 Comparison	figures/figs3.py	X2 to X4; E32 (value in Figure 12)
13 Procedure	figures/fig_procedure.py	none

SCRIPT	SHA-256
ers003_analysis.py	`db85af2de931673fc2a8025a496af3f35dfcd906874b2d4738ec615b6bd59bfb`
figures/fig_mode.py	`f0dd54c592710272127bab42844ae97df7b1e03b06a7d7a0f4f50f6069062325`
figures/fig_board1_raw.py	`b66d3b3ae2dd9c62870e4015977a69dfba15e0327792e5b898b1847b921b2a59`
figures/fig_board1_photos.py	`08effd3fea7be5a0684fab4a6269c929767c5224415e32afae6b8fb2d04f176c`
figures/figs4.py	`fbcf753675545e84d0d86d5beb8fad7f79cc6f1e39e511ab62e59eb8423c7627`
figures/fig_marking.py	`3c94acd212da10011acfdc0ef68a9982c7994f18ce0d615aceb8b3fa77fd89b3`
figures/fig_wiring.py	`260834fd966e45813e84165e88af40cda13c3b7a3bbbfa0dbd41a87041bc9ccd`
figures/figs3.py	`a156b7467e0ce3994550bef25a69535203fda2364cd82f94293f090a0f9a1b30`
figures/fig_procedure.py	`db896f5d38131df2d3deac4afaab9079942f3a2e4c5fc67b05cf209b9a37d974`

8 What cannot be reproduced from files

ITEM	BASIS IN THE REPORT
STATUS (1.8 V on; 0.45 V off, twice) and VDD-EXT (1.81 V; 0.000 V)	Operator multimeter readings, not photographed. Withdrawn readings remain in E15
VIN 3.333 V and VBAT 3.838 V; VBAT-to-VIN diode reading 0.518 V	Operator multimeter readings, not photographed
DIP40CORE-A connections (Figure 8)	UART paths confirmed functionally by the AT exchange. The remaining connections (VIN unconnected; VA, OE and VB supplies) follow the wiring guide given at the bench and the operator's confirmation. The setup photographs show the layout and converter but do not resolve individual connections; no continuity check of the open VIN
Cause of the run 2 high-current events	Operator account: probe slips. Which pins were bridged is not known
Times of the UART unplug and replug	Operator times to the minute; the transients in E23 are plausibly these events, not proven
Waveshare HAT configuration at E12	Inferred from E13, a photograph dated by filename only
Waveshare HAT PWRKEY presses, 15 October	Operator record of the press times, matching the transitions in E31 and E32
Waveshare HAT instrument settings, 15 October	As displayed in the application after each recording (E33, E34)
Waveshare HAT SIM and photograph phases, 15 October	Operator statement: SIM fitted (the holder is not visible in E35 to E37); E35 and E36 show the module off, E37 the module on
Waveshare 3.2 µA figure	As cited in TB-2026-001; no dated copy of the vendor page is held

9 Notes on the record

- The frozen session log (E15) records 26.22 mA for the first on-state interval because its calculation includes the transition second at 433 s. The report uses 26.26 mA, calculated from the 433 complete seconds before the transition, consistent with the stated analysis window. E15 is retained unchanged as the original session record.

- The Track 2 wiring drawing of 11 September 2026 (bg95_pi_level_shift_wiring.png, SHA-256 `b5aa3aa43afe45f508d7cf2e62c7d3e2b5d74370f2646254c98647116e0b233b`) shows the earlier 220 Ω divider on the debug UART and does not describe the 14 October session.

- Because of its size, E21 was processed in two portions split at a CSV line boundary. The per-second summaries of the two portions were combined by summing bin totals and counts.

- The instrument is uncalibrated in source-meter mode at 1.5 to 1.9 mA (ERS-2026-003 §01). Reproduction of these files reproduces the analysis, not the instrument's accuracy.

10 Evidence register

ID	ITEM	SHA-256
E1	RN-PROTO-003 Track 2 Report (BG95-M3 verification). Frozen; SN and IMEI transcription corrected in the campaign record	`7bf2453b6b39d62f61c3dc750e679b47a800c1af7ae134b112545a238d2d797d`
E2	BG-95_module_ON.png (Track 2 on state, 10 min selection)	`6f2a8ead8c3dcb36ea8bf4f79839b1eed392890b308a90cf0ed12bd7b33a52ad`
E3	BG-95_module_off_with_AT_QPOWD.png (Track 2 off state, 10 min selection)	`065275b2268b64dc85aeb9109e3e7c7f00fa7cf4cbc391f44b8f3e3c1ea4cf5e`
E4	BG-95_AT_QPOWD_NO SIM.png (Track 2 on-to-off transition)	`0f1d1f022afc40832a9827e4696b98c3156f9101d2d62f3604c41bb25c9bea92`
E5	1.5mA.png (full view: source meter, 3,800 mV, 100 kS/s, D439F20BA129, v4.4.1)	`8806c70bc8e9cdaa273deca59ac6645a976fcb5acfcd50021d60a93b700a3c05`
E6	ppk2_2.png (reference: "Mode: Ampere meter" logged)	`0337a01c33a95e1bcaf23ebd9a0fba257973ee9541ee6e47b7d0f32fca881686`
E7	RN-PROTO-003 Master Campaign Record (Waveshare HAT PWRKEY state)	`afd55355e7f9fd2025cfb9c14e656c93ca4f79f43ecd758fef3ff02dbc51fdcc`
E8	ERS-2026-002 v1.0 (released)	`b55a4db94d7cdc8786cd1b314bb1f87950ccd832e09e157a19b1fc0060a83767`
E9	RN-BC6-CR-001 (BC-6 D1a and Track 4 closure)	`b8c34fffde01c64dea58b493d9f1182a5c6be56126fc1e8285df10d9442e2175`
E10	TB-2026-001 v1.0 (attribution of 3.2 µA)	`e555bfab9e0831928d0f05c0d5a1f92d8f8f85ae075085244f7d932e640557b3`
E11	RN-STATUS-001 v2.7 (correction record: Waveshare HAT LED state)	`29b6a015fb650091cfe5316a2c054c2b90ab191d64484c4b223de2dff16b0d50`
E12	ppk-20260826T093008.csv (Waveshare HAT off-state raw capture, 10.000 s, 100 kS/s)	`c01b10d9b1b35e9ced3c5d3fe1744ac92d59bd7266699bb1881e69fd9e1057da`
E13	BC4_ppk2-hat-attach_20260826.jpg (ERS-2026-002 repository, data/photos)	`fc34879fd9d9018415c17674b2380469312b22a5e1ab58d0a89636759997e150`
E14	BC4_sim7080g-hat_20260821.jpg (ERS-2026-002 repository, data/photos)	`617cb1afcc4b82058b0894f7d08b1850210496839ebc79c70fe45535f2ea071f`
E15	ERS-2026-003_Bench_Session_Log.md, session ERS003-BENCH-01, 14 Oct 2026 (frozen at release)	`312fcd32c371ac9df1c0bc5f27975347d8f8382a74394df2b37c435087aee538`
E16	A1_1_BG95_marking.jpg (module marking)	`a3d0b02d3fef58ff7024d7b05851ffe6c20c4c8493a75a4408a68f4ff5b42614`
E17	A2_1_ppk2_settings.png (source meter, 3,800 mV, 100 kS/s; log "Mode: Source meter")	`175d133d8a8dd5c77ea3f89c502710dcc1c738d901ee380b9afd3d7b620d899b`
E18	A2_run1_preliminary.ppk2 (on state, 70.6 s, start 20:22:44 WAT)	`0b8d4d606998a7071c43000900744854ec1a9a2ec209f774df7f27b7c2e388b0`
E19	A2_run2.ppk2 (30 min: on, probe-slip events, QPOWD, off; start 20:33:18 WAT)	`3ae971a6932365778db464d8bd3f6fcf123edc60ab11f76e504c92b03826a921`
E20	A2_run3_off.ppk2 (30 min off; start 21:27:09 WAT)	`6e7cd6cd819d685534ad32b6a0db3fe1a243493b2e714ab0c86cfbc4f3275690`

ID	ITEM	SHA-256
E21	A2_run3_off.csv (run 3 raw export, 180,000,001 rows, 5,087,288,923 bytes)	cbc90e1e4c54e96df23a39d40b031cc448d02a9dd496c59cb40db014c1a946c8
E22	A4_uart_disconnect.ppk2 (5 min, UART disconnection)	d5229ae32e3cea104b03d472d841ad8df9cfe38811180f0627ee0f1dd57c0542
E23	A4_uart_disconnect.csv (raw export, 30,000,001 rows)	815c8ee42d31093f38a549c707fb9b54f093802078f381f52bcfbf8d7c6725d9
E24	A2_3_minicom_AT_ATI.png (AT, ATI response)	ae35651725254ce1654dbfa13f11ac553a29fc442388a823282694f07a2932b6
E25	A2_4_minicom_QPOWD.png (AT+QPOWD=1, OK, POWERED DOWN)	702a2560b48dd1d2f76a3be0534f34d573aa6228ce219cdb138ff6e02a453e76
E26	A2_5_run3_selectall.png (= A2_run3_off.png: run 3 selection statistics)	9c327c70a6f32abb5feaf13cbe0cd4f72b003a072bfab77c3c6ee03f4d9fb3bf
E27	A2_run2.csv (last 10 s of run 2, off state)	5eabd903f7cc4a5cd7fc2639e528d5dab86a56405e9160df849ad390233d3177
E28	S0_02_HAT_top.jpg (Waveshare HAT, 14 Oct 2026)	f45736fbd5fe507905783d9b2a9b40724ca2ff1c0bdebc1cd381b0663698f4f2
E29	S0_04_HAT_underside.jpg (Waveshare HAT, 14 Oct 2026)	2ff79c78f8ab1e18b487153a591a39542b5cc556487fc25bc8f0c2182309ad3c
E30	A2_1_run2_screen_t67s.png (run 2 recording: source meter, 3,800 mV, 100 kS/s; log "Mode: Source meter" 20:20:59)	6b85fcfe20e5bc0d2ff326817166b03ef43fd527c2b6d78b39ff107071820871
E31	ppk-20261015T194532.csv (Waveshare HAT recording 1, 30,000,001 rows, 841,789,523 bytes)	f8df6e9e3ff49c11fb56df6fad380c893638f7086cd6ac33b06c0fdbaa1577ff
E32	ppk-20261015T201803.csv (Waveshare HAT recording 2, 30,000,001 rows, 831,741,736 bytes)	50cc29e359a2c0ced3b571b4c325e08faf931c2350aadcb7c6f3d64faad95eaa
E33	HAT_20261015_rec1_screen.png (recording 1 after completion: source meter, 3,800 mV, 100 kS/s)	48386a3b0fe53ba8e790977d8680ca92f06a36f01a120aaecae15cd35e71de9a
E34	HAT_20261015_rec2_screen.png (recording 2 after completion: same settings; log "DUT ON" 21:10:31)	9e45139d70c21948f79b7573a9a84988cd85e94c9f1cc1a618a29abd7930def0
E35	HAT_20261015_off_1.png (module off, STA LED dark)	c89a8294605563af18b2ed9f405743be644a58e702e115748dfef66b53288ab0
E36	HAT_20261015_off_2.png (module off, STA LED dark)	a8dac337ebcc96a7c678652848e5d811918733d17d61e0fa16736159b246a1c1
E37	HAT_20261015_on.jpg (module on, STA LED lit)	56ea46b86004ec5f9d1d7dbba2401be87029a69d2c1d98b15e4053c130ad5c1c
X1	A2_run1_1s_summary.csv (derived from E18)	561278e2545e701dbfc484eaef31b0e670e994bd0bd53f18f7f1e8363b6eae94
X2	A2_run2_1s_summary.csv (derived from E19)	9a4646f7793f91f5d8aa799927e9a243743bea74ec8f3b07cc1611035b259c78
X3	A2_run3_off_1s_summary.csv (derived from E21)	5f9990de14cd21fe194c6fe8780672b847db8c432888961e3690ebde11b3a33d
X4	A4_uart_disconnect_1s_summary.csv (derived from E23)	48b97231140d56aec94c8c7a305fcee54bed126d5cd0cfe0d92c3bba91a94af8

ID	ITEM	SHA-256
D1	Quectel BG95 Series HD V1.5, 2022-10-09, Table 42, footnote 24	`f6b428e09a3b9b227dd6667ae83686db78ea99bd7d00e0785d99ff19d0ce79d9`
D2	SIMCom SIM7080G HD V1.04, cover date 2020-09-27 (own version history: 2020-08-11), Tables 6, 30, 33	`efd0834d62635be27b7f1c1eae420add04c24bfa8c824fa91bc39892c03ec81a`
D3	SIMCom SIM7080G HD V1.03, cover date 2019-12-13 (V1.04 version history: 2019-12-25), Tables 6, 31	`d38e91011c3c3a90b49579e405ad2ee6bfa003694d90ba83ea8e68f21a0312c2`
D4	Waveshare SIM7080G HAT product page (PSM 3.2 µA), as cited in TB-2026-001 (E10)	[no dated copy held]
D5	Waveshare SIM7080X Cat-M/NB-IoT HAT schematic (Altium, 2020-01-19)	`db53a98fae4c846202f70d6569a280a3b9a0fc1b0e1971f4d3e3a5e8e46fa899`
D6	DIP40CORE-A BG95 (20230417) schematic	`885836cc01f4f4cf7b256d63fc370316e1b3d891d49417425cca361d2f69d404`
D7	BG95 40PIN DIP PCBA Specification (Release 2023-04-06)	`3a4e90cbefd6a0888acbcb9bd8a27be8fba35b13d0ed86d6fd267aae6cf06e27`
D8	SIMCom SIM7080G HD V1.05, 2023-06-14, Table 7	`e5058ce39bb6ae783a237c0fec7ee3efe1d20fc51ef73c2c6756b2cb8ef76761`

E2 to E5 are Track 2 screenshots saved 12 September 2026. E15 to E30 are from bench session ERS003-BENCH-01, 14 October 2026; E18 to E23 and E27 were hashed in place on the operator\'s PC. E31 to E37 are from the Waveshare HAT re-measurement of 15 October 2026; E31 and E32 were hashed in place on the operator\'s PC, and E33 to E37 are the copies supplied by the operator. X1 to X4 were computed from the raw files (method in section 5; notes in section 9).

Analysis script SHA-256: `db85af2de931673fc2a8025a496af3f35dfcd906874b2d4738ec615b6bd59bfb`